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1)#include<stdio.h>
int main()
{
int i,n,*p;
scanf("%d",&n);
int arr[n];
p=arr;
for(i=0;i<n;i++){
scanf("%d", (p+i));
}
for(i=0;i<n;i++)
printf("%d ",*(p+i));
for(i=0;i<n;i++)
*(p+i)=*(p+i)+10;
printf("
");
for(i=0;i<n;i++)
printf("%d ",*(p+i));
}
Output:
4
10
20
30
40
10 20 30 40
20 30 40 50

2)#include<stdio.h>
int main(){
char str[100],*ptr;
int i,len = 0;
ptr=str;
scanf("%s",str);
for(i=0;*(ptr+i)!='';i++)
len++;
printf("%d", len);
}
Output:
bharath
7

3)#include<stdio.h>
void sumprod(int *a, int *b){
printf("Sum:%d
",*a+*b);
printf("Product:%d",*a* *b);
}
int main(){
```

```
int x,y;
scanf("%d",&x);
scanf("%d",&y);
sumprod(&x,&y);
}
Output:
4
7
Sum:11
Product:28

4)#include<stdio.h>
#include<stdlib.h>
#include<string.h>
struct Array
{
int num;
char str[20];
};
int main()
{
int n,i;
scanf("%d",&n);
struct Array *ptr;
ptr=(struct Array*)malloc(n*sizeof(struct Array));
for(i=0;i<n;i++)
{
printf("Enter num:");
scanf("%d",&(ptr+i)->num);
printf("Enter str:");
scanf("%s", (ptr+i)->str);
}
printf("
Before modification:
");
for(i=0;i<n;i++)
{
printf("%d ",(ptr+i)->num);
printf("%s
",(ptr+i)->str);
}
if(n >=5)
{
(ptr+3)->num = 5;
strcpy((ptr+4)->str, "Hello");
}
printf("
After modification:
");
for(i=0;i<n;i++)
{
printf("%d ",(ptr+i)->num);
printf("%s
```

```
",(ptr+i)->str);
}
free(ptr);
return 0;
}
Output:
5
Enter num:100
Enter str:alex
Enter num:200
Enter str:john
Enter num:300
Enter str:kiran
Enter num:400
Enter str:dave
Enter num:500
Enter str:rahul

Before modification:
100 alex
200 john
300 kiran
400 dave
500 rahul

After modification:
100 alex
200 john
300 kiran
5 dave
500 Hello


5)#include<stdio.h>
int math(int a, int b){
return a+b;
}
int math2(int a, int b){
return a*b;
}
int main(){
int (*f1)(int, int);
int (*f2)(int, int);
f1= math;
f2= math2;
printf("Sum:%d
",(*f1)(2,3));
printf("Multiply:%d", (*f2)(2,3));
}
Output:
Sum:5
Multiply:6
```

```
6)#include<stdio.h>
int main()
{
int num, *ptr, **pptr;
ptr = &num;
pptr = &ptr;
scanf("%d",&num);
printf("Value of num: %d
", num);
printf("Value using ptr: %d
",*ptr);
printf("Value using pptr: %d
",**pptr);
printf("Address of num: %p
", (void*)&num);
printf("Address stored in ptr: %p
", (void*)ptr);
printf("Address of ptr: %p
", (void*)&ptr);
printf("Address stored in pptr: %p
", (void*)pptr);
return 0;
}
Output:
7
Value of num: 7
Value using ptr: 7
Value using pptr: 7
Address of num: 0x7ffd9a41b2c4
Address stored in ptr: 0x7ffd9a41b2c4
Address of ptr: 0x7ffd9a41b2b8
Address stored in pptr: 0x7ffd9a41b2b8

7)
#include<stdio.h>
int smallest(float a[], int k, int n)
{
int i, pos = k;
float small = a[k];
for(i=k+1; i<n; i++)
{
if(a[i] < small)
{
small = a[i];
pos = i;
}
}
return pos;
}
void sel_sort(float a[], int n)
{
int k, pos;
float temp;
```

```
for(k=0; k < n - 1; k++)
{
pos = smallest(a, k, n);
temp = a[k];
a[k] = a[pos];
a[pos] = temp;
}
}
int main()
{
int i, n;
scanf("%d",&n);
float a[n];
for(i = 0; i < n; i++)
scanf("%f",&a[i]);
sel_sort(a,n);
printf("Sorted array:
");
for(i = 0; i < n; i++)
printf("%.2f ",a[i]);
return 0;
}
```

Output:

```
5
4.2 1.8 9.5 3.1 0.7
Sorted array:
0.70 1.80 3.10 4.20 9.50
```

8)

```
#include<stdio.h>
#include<string.h>
struct Student
{
int id;
char name[20];
char grade;
};
void sort(struct Student a[], int n, int choice)
{
int i,j;
struct Student temp;
for(i=0;i<n-1;i++)
{
for(j=i+1;j<n;j++)
{
if((choice==1 && a[i].id>a[j].id) ||
(choice==2 && strcmp(a[i].name, a[j].name)>0) ||
(choice==3 && a[i].grade>a[j].grade))
{
temp = a[i];
a[i]=a[j];
a[j]=temp;
}
```

```
}
}
}
int main()
{
int i,n,choice;
scanf("%d",&n);
struct Student a[n];
for(i=0;i<n;i++)
scanf("%d %s %c", &a[i].id, a[i].name,&a[i].grade);
printf("Select sorting criteria:
");
printf("1-ID 2-Name 3-Grade
");
scanf("%d", &choice);
sort(a,n,choice);
for(i=0;i<n;i++)
printf("%d %s %c
", a[i].id,a[i].name,a[i].grade);
return 0;
}
Output:
4
105 bharath B
102 arun A
108 kumar C
101 divya A
Select sorting criteria:
1-ID 2-Name 3-Grade
1
101 divya A
102 arun A
105 bharath B
108 kumar C

9)#include<stdio.h>
void radix(int a[], int n)
{
int bucket[10][20];
int bucket_count[10];
int i,j,k, digit, div = 1,max=a[0];
for(i=1;i<n;i++)
if(a[i]>max)
max = a[i];
while(max/div>0)
{
for(i=0;i<10;i++)
bucket_count[i]=0;
for(i=0;i<n;i++)
{
digit=(a[i]/div)%10;
bucket[digit][bucket_count[digit]] = a[i];
bucket_count[digit]++;
```

```
}
k=0;
for(i=0;i<10;i++)
{
for(j=0; j<bucket_count[i];j++)
{
a[k]=bucket[i][j];
k++;
}
}
div=div*10;
}
}
int main()
{
int a[20], pos[20],neg[20];
int n,i,p = 0, k = 0, j = 0;
scanf("%d",&n);
for(i=0;i<n;i++)
scanf("%d",&a[i]);
for(i=0;i<n;i++)
{
if(a[i]<0)
neg[k++]=-a[i];
else
pos[p++]=a[i];
}
radix(neg,k);
radix(pos,p);
for(i=k-1;i>=0;i--)
a[j++]=-neg[i];
for(i=0;i<p;i++)
a[j++]=pos[i];
printf("Sorted array:
");
for(i=0;i<n;i++)
printf("%d ",a[i]);
return 0;
}
Output:
6
-125 450 -820 215 -305 670
Sorted array:
-820 -305 -125 215 450 670

10)
#include<stdio.h>
int main()
{
int a[20], b[20], n, i;
int *p, *q;
scanf("%d",&n);
for(i=0;i<n;i++)
```

```
scanf("%d",&a[i]);
p=a;
q=b;
for(i=0;i<n;i++)
{
*(q+i)=*(p+i);
}
printf("Copied array:
");
for(i=0;i<n;i++)
printf("%d ",*(q+i));
return 0;
}
Output:
5
15 25 35 45 55
Copied array:
15 25 35 45 55
```